Wild life - Animals living in their natural habitat, rare species of plants and animals

- For balance of nature
- As genetic resource
- Economic
- Recreation
- Education

Causes of wildlife depletion

- Deforestation
- Mining
- Hunting
- Poaching
- Hydroelectric projects
- Pollution
- Natural calamities
- Man animal conflict
- Pesticides and fertilizers

During last 2000 years, 106 species of animal and 139 species of birds have become extinct
Plants 450 species BSI and animals 2000 species ZSI areendangered
Mammals – lion tailed monkey, nilgirilangur, wolf, jackal, bear, lion, tiger, leopard, rhino, swamp deer, musk deer, Indian bison, gibbon, tiger civet, elephant and dolphin
Birds – swan, sea eagle, great indian bustard, hornbill and peacock
Reptiles – green turtle, crocodile and python

Methods of conservation

- Knowledge of wildlife
- Protective laws
- No hunting and poaching
- Habitat improvement
- Epidemic control
- Census
- Creation of more national parks and wild life sanctuaries

Our Earth has more than 20 million species; only 1.7 million species are described till date
In plants – Angiosperms and in animals – Arthropods are the largest group.

- Pisces – 21723
- Amphibian – 5150
- Reptilian – 5817
- Aves – 9026
- Mammalia – 4629
- Arthropods – 987949

India is one of the 12 mega diversity center of the world, India has only 2.4% but 7.31 % of global animal diversity

- Arthropods – 68389
- Pisces – 2546
- Amphibian – 209
- Reptilian – 456
- Aves – 1232
- Mammals – 390

Endemic species of India – the species which occur in small area are known as endemic, Biodiversity in India show high level of endemism
33 % of plants are endemic to India

- 110 species Amphibia
- 214 species reptilian
- 69 species aves
- 38 species mammals

Introduction to Animal Classification

All living things are divided into seven categories using Latin terms that can be understood worldwide. They progressively get smaller and more numerous, and each contains less living things. Classification of living things is used to help identify different animals and to group them together with their relatives.

The first and largest category is the Kingdom. To date there are five kingdoms: Animalia, which is made up of animals; Plantae, which is made up of plants; Protista, which is made up of protists (single-celled creatures invisible to the human eye); Fungi, which is made up of mushrooms, mold, yeast, lichen, etc; and Monera, which is made up of the three types of bacteria.

The next category is the Phylum. There are several phyla within each kingdom. The phyla start to break the animals (or plants, fungi, etc) into smaller and more recognizable groups. The best known phylum is Chordata, which contains all animals with backbones (fish, birds, mammals, reptiles, amphibians). There is also Arthropoda (insects, spiders, crustaceans); Mollusca (snails, squid, clam); Annelida (segmented worms); Echinodermata (starfish, sea urchins) and many, many more.

The next category that makes up the phyla is the Class. The class breaks up animals into even more familiar groups. For example, the phylum Chordata is broken down into several classes, including Aves (birds), Reptilia (reptiles), Amphibia (amphibians), Mammalia (mammals) and several others.

The next category is the Order. Each class is made up of one or more orders. Mammalia can be broken down into Rodentia (mice, rats), Primates (Old- and New-World monkeys), Chiroptera (bats), Insectivora (shrews, moles), Carnivora (dogs, cats, weasels), Perissodactyla (horses, zebras), Artiodactyla (cows), Proboscidea (elephants) and many more.

Orders can then be broken down into Families. The order Carnivora can be broken down into Canidae (dogs), Felidae (cats), Ursidae (bears), Hyaenidae (hyaenas, aardwolves), Mustelidae (weasels, wolverines), and many more.

The next category is the Genus. The family Felidae, for example, can be broken down into Acinonyx (cheetah), Panthera (lion, tiger), Neofelis (clouded leopard) and Felis (domestic cats).

Finally, the genus is broken down into the Species. The genus Panthera can be broken down to include Panthera leo (lion) and Panthera tigris (tiger). Note that the genus is placed in front of the species.

To complicate things, each category can, in some cases, be broken down into sub-categories. A sub-category is made when there are some animals that are basically the same but live in different areas and behave differently from each other or have some general differences in the anatomy. A prime example is phylum Chordata, class Reptilia, order Squamata. The order Squamata consists of both lizards and snakes because they are basically the same. However, snakes have some physical differences, mainly the lack of legs (although some lizards also lack legs). This warrants the creation of two sub-orders: Serpentes (snakes) and Sauria (lizards). These are then broken down into their own families.

THE CHORDATES

Chordates are characterised by a rod-like notochord and a hollow nerve cord on the dorsal side of the gut, and pharynx being perforated with gill slits for respiration. Vertebrates have a vertebral column that replaces the notochord, cephalization of the nervous system, a ventral heart, a post-anal tail and division of the coelom into chambers.

Group CRANIATA (=EUCHORDATA)

(Chordates with skull, with 54,000 species of true chordates)

Subphylum VERTEBRATA, chordates with head, brain and vertebral column.

Class MAMMALIA

Mammals evolved in the late Triassic, the time dinosaurs first appeared and diversified greatly following the extinction of dinosaurs during the Coenozoic. Characteristics include hairs for protection and from heat loss; mammary glands; heterodont teeth; endothermy; 4 chambered heart, a large cerebral cortex, three distinctive bones: incus, malleus and stapes in the middle ear, a diaphragm for breathing, limbs attached under the body, dicondylic skull and acoelous vertebrae. The class Mammalia is classified into three subclasses, 28 Orders, 161 Families, 747 Genera and 4939 Species

General characteristics of mammals

- Secrete milk
- Warm blooded
- Have hairs on body, Furred
- Unique jaw structure, movable
- Larger brains
- Have back bone
- Ability to suckle young ones

- Skin has sweat glands
- Metacarpal and metatarsal fuse to form common bones
- Neck and tail
- Dentition is heterodont
- Separate openings for alimentary and urogenital canal
- Heart is 4 chambered
- Non nucleated RBC
- viviparous

Subclass PROTOTHERIA, (Egg-laying mammals)
Order Monotremata includes 2 families of mammals that lay eggs with leathery shells and nourish the young with milk from primitive mammary glands. They possess a cloaca like reptiles, have no urinary bladder but possess hairs.Found in Australia and New Zealand, No teeth and true mammary glands.Most primitive of the mammals
Only one order – monotremata – echidna, platypus
Egg layers and exude milk from pores of skin

Subclass METATHERIA, (Marsupial mammals)

Includes about 300 species grouped under about 240 genera and 15 families, whose young are born in immature state and then reared in pouches called marsupium. Found in Australia and South America. Opossum has now spread to North America.

Order Marsupialia (Body covered with fur; pinna well developed; female with marsupium; tympanic bulla absent; diaphragm and seven cervical vertebrae are present and the marsupial bone (epipubis) present), marsupials that have brief gestation period after which the embryo develops in a pouch. They have prolonged lactation and parental care. Marsupials include: opossum, kangaroo, koala, Tasmanian devil, wombat, etc.

Kangaroos and opossums, in kangaroos the hind limps are much longer & stronger than forelimbs helps in jumping, gestation period is 28-38 days, single young ones, life span is 12-18 years

Subclass EUTHERIA (Placental mammals)
16 orders which include 4700 species of placental mammals those are truly viviparous, with a placenta for gas and nutrient exchange between the mother and foetus. They also have true mammary glands, with chorio-allantoic placenta and in which complete development takes place in uterus.
Insectivora – insect eaters – hedgehog & shrews, small active secretive mammals, have ability to curl upon into a ball, gestation period 30 days, litter size 4-7
Chiroptera– 1000 spp - bats – find thier path by echolocation, high frequency sounds
Megachiroptera – frugivorous
Microchiroptera - insectivorous
Primates - lemurs, monkeys, apes and man
Highest order of mammals, brain well developed, hands &feets are made for grasping objects, double bone in fore arm helps in free movement of wrist, toes are long and flexible, gibbons have longer arms & weaker hind quarter whereas langurs have longer legs than arms, tail helps in balance, monkeys are good swimmers, hands are the prehensile organs, langurs are herbivorous and macaques are omnivorous, lemurs are nocturnal, have got well developed vision & hearing, alert & agility, live in collective groups, give alarm calls, live in tropical climate, monkeys express pleasure, anger and fear, fur picking is the emotional caress for

powerful bondage between them, these are intelligent animals and look after their young ones for long time,
Indian – apes, gibbons, monkeys and lemurs
Exotic apes – no tail and use arms
Africa – gorilla (largest primates) and chimpanzee
Sumatra – orangutans, 1.5 mt height and 90 kg weight
Chimpanzee – Africa, 90 kg weight, maturity age 7-10 yrs, gestation period 230-285 days, builds nest on trees, life span 60 yrs
Hoolock Gibbons –the ape of India - assam, tailless and have long arms, stand erect, 6.8 kg weight and 1 mt height, live in small groups, usually gives birth in Dec & Mar, life span of 25-30 yrs
Macaque – sturdy and have cheek pouch, rhesus-north, bonnet-south and have longer tail, lion tailed macaque – Western Ghats, powerful jaw, gestation period of 6 months, life span of 25 yrs
Langur – slim and pouched stomach,estrus cycle 30 days, gestation period 6 months, live in troops, life span 30 yrs, common langur – entire country, nilgirilangur – kerala,TN and Karnataka
Lemurs – are nocturnal, in India – slow loris found in NE and slender loris in south India
Dermoptera – winged skin - flying lemurs – leaf eaters
Pholidota – pangolins – Africa & Asia, pointed head, long tail & tongue, feed on termites and ants, horny body scales, powerful limbs, absence teeth, 1-3 young ones, gestation period 140 days, can roll into balls, life span 14 yrs
Proboscidea – elephants –largest terrestrial animals 4-7 tonnes, massive head, small eyes, tusk & trunk, movable ears, GP 21 months, life span is 60-70 yrs, senior female in the herd is leader, African & Asian, simple stomach and no gall bladder, abdominal testis, estrus cycle 120-130 days, gestation period 20-22 months, teats are between anterior limbs, CI – 4 years upto 60 yr of age, intelligent animal, strong liking for salt and minerals, musth swelling of temporal glands, aggressive behavior and high level of testosterone, life span of 70 yr,
Carnivora – meat eating mammals, long canine teeth, 4-5 clawed toes, mostly ground living, aquatic- sea otter, arboreal-civet, digitigrade- bears, plantigrade- dogs, smallest carnivore- weasel, largest carnivore- polar bear, many have os penis, solitary life

Felidae – 37 spp, hearing & smell well developed, nocturnal and are good swimmers, tigers, lion, leopards, cheetah, jaguar, Cheetha- Acinonyxjubatus - fastest land animal, 50-80 kg, have tear stripe, larger nasal passage, seasonally polyestrus, no regular breeding pattern, gestation period 90 days, 10-12 yr life span,

Lion – panther leo – Asiatic and African, open grasslands, 240 kg, males have mane, 3 cubs, gestation period 92-119 days, most social animals, 2-4 dominant male+female+cubs, urine markings, patrolling or roaring, hunt during day, lion normally eat the hind quarter, 15 yr life span

Leopard – panther pardus- Africa and asia, widespread species, highly adaptable and hunting behavior, the black panther- NE & western ghats, 30-90 kg, gestation period 90 days, crossed with lions leopons, least social animals, hunt singly during night, take the prey on tree, 12-25 yr life

Tiger – panther tigris – asia, 384 kg weight, male have few stripes than female, longer hind limbs, cubs are born usually in april& November, esrus 3-10 days, mate 100 times, 3 young, gestation period 103-115 days, interval between pregnancy is 3 years, good swimmers, do not climb, leopards and tigers avoid each other, tiger drinks frequently between meals, 15 yr life span

Snow leopard – panther uncial – Himalayan range, 23-41 kg weight, gestation period 93-105 days, litter size 1-4, nocturnal, solitary, 15 yr life span

Viveridea– 72 spp, long thin body and long tail, tiger civet, Malabar civet and Himalayan palm civet, they are omnivorous, 90 cm long and 5 kg weight

Hynaetidea – hyaena – nocturnal, ability to crush bones and have got strong jaws, found in asia and Africa, in India stripped hyaena are found, 30-40 kg, forequarter is heavier, longer forelimbs, bushy tail, powerful jaws, dorsal mane, breed throughout year, litter size 1-6, gestation period 90-110 days, master scavengers, hunt in packs, life span 20 years

Canidea– 37 spp – wolves, jackles and foxes, pointed muzzle, bushy tail, family hunting, anal scent glands, erected and pointed ear, os penis, hunt in pack, 18-20 kg, gestation period 60 days, 15 year life span

Ursidea –Asia, America and Europe- 9 spp – polar bear, brown bear and sloth bear - omnivorous, can climb and dig, eat termites, gestation period of 210 days, life span 40 years,

Aluropodidea– red panda and gaint panda, red panda – Aitrusfulgens – Himalayan raccoon, 3-5 kg, thick fur, reddish brown, pointed snout, gestation period 90-105 days, very gentle, life span 14 year

Mustilidea – 67 spp –fresh and salt water , long body and short limbs, musk is secreted by glands, solitary - otters, martens, weasles and ratels – small thin brown body, nocturnal and burrows hole, otter – amphibious, 12 kg, webbed neck, black brown, gestation period 60 days, docile, life span 12 yrs

Rodentia– rodents – 1702 spp – short limbs, nocturnal, small in size, life span is 2 years, for biomedical research, role in spread of disease plague and thaeleria, mice, rats, guinea pigs, hamsters, porcupines and squirrels

Lagomorpha –58 spp - rabbits and hare – soft fur, long ear, slit like nostrils, 2 pair of incisor in upper jaw, in rabbits ovulation takes only after coitus, gestation period 30 days, nocturnal, anal glands odour, nest building activity, life span of 13 yrs

Ungulates –

Perrissodactyla – odd no of toes, large caecum, inguinal mammary glands – ass, horses, tapirs and rhino, Rhino- nose+horn, 1500 kg second heaviest terrestrial animal, one horned & two horned, found in Assam and WB, open grasslands, stout limbs, thick skin, horns grow continuously, penis backwards retracted, estus 40-60 days, gestation period 15-16 months, solitary animal, male marks territory by urine and faeces, dung piles

Artiodactyla – even no of toes - bovidae, cervidae, suidae

Wild Bovids – bison, yak, mithun, Gaur, Bautang, wild sheep/goat

Yak – grunting ox, Tibet, dark brown, short legs, drooping heads,

Mithun – NE India, black color with white extrimities, short tail, docile, large dewlap

Gaur – largest among cattle, small herds, lateral display

Camel – brown color, male protrudes red pouch during rutting

Giraffe – Africa, tallest, brown with dark spots, forelimbs are longer, both sexes are horned, thin and mobile upper lip, long neck, large eyes, small groups

Hippopotamus – Africa, lakes & rivers, barrel shape body, smooth skin, short legs, canine teeth, parturition in water, territory marked by faeces

Antelopes – Swift moving, Africa is the land of antelopes - black buck, nilgai, chinkara and chowsingha – have horns

Nilgai – dark bluish grey, dark mane, tuft of hair on throat, small groups, smooth upright horns, have habit of defecating at one regular spot

Blackbuck – Krishnamruga, have black large spirally twisted horns and females are hornless, prominent glands below eyes

Chinkara – Chestnut color, both sexes are horned, nocturnal

Chowsingha – light brown color, male have two pairs of horns, in herds, regular water drinking habit

Cervidae – all deers – male have got antlers which shed periodically, elongated body and slender legs, short tail, fissure in skull infront of eye socket, no gall bladder, musk deer – male and female do not have antlers whereas reindeer both have antlers, musk deer have upper canines

Swampdeer/barasingha – UP,MP,WB,Assam - gregarious animals, mane present in males

Sambar – largest deer of SE Asia, large infraorbital glands, branched antlers, dark brown color, bushy tail

Cheetal/ Spotted deer – interdigital glands, Fawn color with white spots, symbiotic association with langurs

Muntjack/barking deer – Rib faced deer, male have tusk, cries like dog, dark brown color

Musk deer – females are heavier, golden color, long hind limbs, long canine teeth, small & naked tail, glands between genital & umbilicus, shy & solitary animals

Mouse deer/ Chevrotain – Asia & Africa, related to pigs, solitary and nocturnal, light stripes on the body, copulation for long time, gall bladder present

Suidae– 9 spp – wild hog and pigmy hog

Exotic mega mammals – African elephant, Grizzly bear, Giraffe, Hippopotamus, Rhino, Zebra

Marine mammals

Pinnipedia – sea lion and sea cow

Cetacean – whale, dolphin

Sirenia – dugongs, manatees

CLASSIFICATION OF REPTILES

Found more in temperate and warm regions, helps in control insects and rodents,evolved from primitive amphibians, have lungs, no gills and have horny scales, cold blooded, skin is dry, rough, non-glandular with epidermal scales, sternum is highly developed, heart has impartially divided compartments, absence of hairs and mammary glands, excrete uric acid, tongue protruded, mostly oviparous, 7800 species, turtles, crocodiles, lizards and snakes, they have internal fertilization and produce large cleioid eggs with leathery shells and are ectotherms. The upper part of the skull of reptiles is modified giving the reptiles a far more efficient and powerful jaw action and making the skull light. Majority are carnivorous (snakes) whereas marine turtle & land turtle are herbivorous, Most snakes are egg laying except vipers, teeth of reptiles are replaced throughout life, well developed vision, poor CNS, are poikilothermic, energy requirement is less so can go without feed for long time, python & pit vipers hunt by heat sensors, forked tongue is the organ of smell, have got no ear

Key features of Class REPTILIA

- Body varied in shape, covered with horny epidermal scales, sometimes with dermal plates; integument with few glands.
- Paired limbs, usually with five toes with claws, adapted for climbing, running or paddling; limbs absent in snakes and some lizards.

- Skeleton well ossified; ribs with sternum except in snakes, forming a complete thoracic basket; skull with single occipital condyle.
- Respiration by lungs.
- Three-chambered heart, except in crocodiles which have four-chambered heart.
- Metanephric kidney; uric acid is the main nitrogenous waste.
- Ectothermic animals.
- Nervous system with primitive brain, spinal cord dominant. There are 12 pairs of cranial nerves.
- Sexes separate; fertilization internal, hemipenis as copulatory organ.
- Eggs covered with calcareous or leathery shells. Extra embryonic membranes, amnion, chorion, yolk sac and allantois are present during embryonic life.

Order Chelonia

Tortoises and turtles do not have teeth but possess horny beaks. Tortoises are usually herbivorous while sea and fresh water turtles are omnivorous. Body is covered with a shell consisting of two parts – the dorsal carapace and the ventral plastron, which are connected by bridges between front and hind legs. The ribs and backbone are fused with the carapace, slow moving, non-aggressive, no teeth,toes are webbed, breeding in water, female lays eggs on land, eggs are hatched by heat of soil, retractable head and neck, longest living reptiles about 100 yrs.

Turtles - found more in warmer region, excellent eye sight, aquatic turtles are carnivorous, land are herbivorous, Olive ridley turtle – indo-pacific ocean, brown in color, lay 40-150 eggs, 1 in 1000 believed to survive, mass nesting behavior

Tortoise – temperate and tropical region, high dome, very slow moving, clawed limbs,lay leathery eggs, eggs at low temperature will be males and high temperature will be female as opposed in crocodiles

Terrapin – freshwater tortoise, in ganga basin, delicacy meat

Order Rhyncocephalia

The order contains only two species that live on some islands off the coast of New Zealand. They look like lizards but there are differences that set the tuatara apart from lizards. The tuatara spends daytimes in burrows. It comes out in the evening to feed on insects and other invertebrates.

Order Crocodilia

This order includes alligators, caimans, crocodiles and gharials that are found in and near water in warmer areas of the world. They eat fish, birds, turtles, and mammals.

Members of the crocodile group have legs and feet designed for walking on land and a strong flattened tail used for swimming. have hard scales, short and powerful legs, 5 toes on fore and 4 toes on hind limb, jaw has sharp teeth, well developed smell, sight and hearing, copulation in water, female mates only once in year, lays hard shelled 20-100 eggs,these are decendents of dinosorous, found in rivers and estuaries, eggs are hard, fast swimmers, can hold breath for 1 hr, have sharp teeth which are replaceable,live for 50-60 yrs, sharp vision even during night, muscles for closing the jaw are well developed, 5-150 eggs are laid, marsh crocodile- in india, salt water crocodile- in NE India, Gharials – in north India which have got narrow snout with bulged tip.

The three groups are distinguished from one another by the shape of their heads. Alligators have a broad, rounded snout; while the crocodiles have a triangular head with a more pointed snout and gharials have a very long and narrow snout

Mugger/fresh water crocodile – 200 kg, broad snout, 4 occipital scales behind head, toes are webbed
Salt water crocodile – 5 mt long, 400 kg, longer snout, no post occipital scales,
Alligators – North America and China
Gharial – Ganga, 12-15 feet long, dark olive color, long narrow snout, male has protuberance on snout, lays 40 eggs in pit.

Order Squamata

The order includes Lizards and snakes, which are creepers and inhabit a variety of habitats. Snakes are carnivorous but lizards eat a variety of foods including plants and insects. Snakes have descended from lizards and there are many similarities between them. Some characteristics that distinguish snakes from lizards are:

- Snakes do not have eyelids but lizards have.
- Snakes usually have one row of scales on the belly; lizards have many.
- Snakes do not have legs; most lizards have legs.
- Snakes have jaw bones that disarticulate allowing them to swallow large objects. Lizard jaw bones do not disarticulate.

Lizards – have eyelids, ear opening, 2800spp, lay eggs, home lizards, garden lizards, Chameleons, Monitor lizards, Komodo Dragon (Indonesia)

Snakes –found more in tropics, evolved from lizards, 2750 spp, jaw bones can move apart which helps to engulf prey, have long body without limbs & eyelids, moult skin several times (2 MONTHS), outer ear is absent, no diaphragm, forked long tongue, absence of urinary bladder, about 300 venomous spp – more than 50 are dangerous to humans
Python – 27 spp, massively build, grey color with flattened head, lethargic and slow moving, swims and climbs, kill by squeezing body, Anaconda, Indian Rock & Indian Reticulate
Reticulate python – largest – 8.5 to 10 mt long

King Cobra *(Ophiophagus hannah)*

The third largest snake in India, the king cobra is also one of the most courageous, poisonous and aggressive snakes. Their hood is relatively less inflatable than that of the Indian Cobra. The body is blackish brown with lighter bands running throughout its entire length. King cobra has a voracious appetite and their staple diet is snakes but they also eat lizards and rats. Mating has been observed in March, with copulation lasting for about an hour. The eggs are laid 5-6 weeks after mating. A maximum of 51 eggs have been recorded. King cobras can reach 18 feet (5.5 meters) in length, making them the longest of all venomous snakes. When confronted, they can raise up to one-third of their bodies straight off the ground and still move forward to attack. They will also flare out their iconic hoods and emit a bone-chilling hiss that sounds almost like a growling dog. Their venom is not the most potent among venomous snakes, but the amount of neurotoxin they can deliver in a single bite is up to 7 ml that is enough to kill 20 people or an elephant.

Indian Cobra *(Naja naja)*

One of the most poisonous snakes found in India, the cobra has the most distinctive features as its hood. There are three races mainly theIndian Spectacled Cobra *(Najanaja naja)*, the mono-ocellate cobra *(Naja naja kaouthia)* and the black cobra. Cobra feed principally on rats, frogs and toads, but can also take in lizards and snakes and also earthworm, ants, cockroaches and eggs. Mating occurs in January and 12-22 eggs are laid in April-May, which hatch within 45 to 69 days.

Bamboo pit viper *(Trimersurus gramineus)*

All vipers give birth to live young which hatch from eggs inside the mother's uterus. They can be distinguished from all other groups of snakes by the presence of the loreal pit (heat sensitive pit enabling the snake to locate and capture its prey). This beautiful but poisonous snake has a glossy green back and yellowish white bottom. It is usually sluggish during the day but when provoked it anchors itself firmly by its prehensile tail with an open mouth. It prefers hilly forests and bamboo forests.

Russell's viper *(Daboia =Vipera russelli)*

This is a medium to large-sized snake with a characteristic bright pattern on its body. Found all over India, both in the plains and hills up to an elevation of about 3,000 mts. This snake grows to a maximum length of 166 cm. The average length is about 120 cm on the mainland, although island populations do not attain this size. The head is flattened, triangular and distinct from the neck. The snout is blunt, rounded and raised. The nostrils are large, in the middle of a large, single nasal scale. The lower edge of the nasal touches the naso-rostral. The supra-nasal has a strong crescent shape and separates the nasal from the naso-rostral on the anterior. The rostral is as broad as it is high. The body is stout, the cross-section of which is rounded to cylindrical. The dorsal scales are strongly keeled; only the lower row is smooth. It feeds primarily on rodents, especially murid species. However, they will eat just about anything, including rats, mice shrews, squirrels, domestic cats, land crabs, scorpions and other arthropods. Adults are reported to be persistently slow and sluggish unless pushed beyond a certain limit, after which they becomes fierce and aggressive. Juveniles, on the other hand, are generally more active and will bite with minimal provocation.

Saw-scaled Viper *(Echis carinatus)*

A small-sized snake found all over India, usually in the plains. They may occur in areas as high as 2,000 mts. in the north-western Himalayas. Size ranges between 38 and 80 cm in length, but usually no more than 60 cm. Head distinct from neck, snout very short and rounded. The nostril between three shields and head covered with small keeled scales, among which an enlarged supra ocular is sometimes present. They move about mainly by side winding: a method at which they are considerably proficient and alarmingly quick. It feeds on rodents, lizards, frogs, and a variety of arthropods, such as scorpions, centipedes and large insects. The population in India is ovoviviparous. In northern India, mating takes place in the winter with live young being born from April through August.

Himalayan pit viper *(Gloydius himalayanus)*

It is a venomous pit viper species found along the southern slopes of the Himalayas in Pakistan, India and Nepal. Medium-sized snake, with distinct elongated head covered with large symmetrical scales; a distinct pit between eye and nostril. Dorsum light brown, gray to dark brown. A median series of dark brown blotches, alternating with lateral row of spots. A broad dark band from eye to the angle of mouth. Supralabials light with dark mottling. Ventrum light gray with dark clouding and fine spotting. This snake frequents rocky wooded hill sides, where it lives in caverns and crevices among rocks, hibernates in winter from October to April. Basks in bright sunny winter days, habitually sluggish. Feeds mainly on skinks and other mountain lizards.

Common Krait *(Bungarus caeruleus)*

Found almost all over India, they are nocturnal in habit. Commonly found in northeast India, Bihar, Orissa, Madhya Pradesh, Andhra Pradesh and Uttar Pradesh up to an elevation of 1500 mts. It is easily identified by its triangular body cross-section and the marked vertebral ridge consisting of enlarged hexagonal vertebral shields along the middle of the dorsal side. The head is not distinct from neck. The eye is black. It has arrow-head like yellow markings on its black head and has yellow lips, lore, chin and throat.

Banded Krait or *(Bungarus fasciatus)*

They are medium to large-sized snake with prominent yellow and black bands on the body. The banded krait has been recorded to grow up to a length of 2 metres in M.P., Bihar, Orissa, U.P. and Assam. Sometimes there is a black spot on the head. It feeds on other snakes.

Sleder Coral Snake *(Callophis melanurus)*

A small slender snake found in most parts of the country except parts of central and northeast India, Bihar, Orissa, Madhya pradesh, Andhra Pradesh and Uttar Pradesh up to an Elevation of 1500 mts. A small and slender snake that has uniform pale brown body colouration. This snake can be identified by its non-evident neck, blunt snout, widely spaced nostrils which nearly are as wide apart as the eyes and small Supra-ocular compared to the Frontal. Its head is mottled and dotted with black from which its specie name has been derived. The tip of its tail may also have a similar colouration and it could serve to act as a decoy. Some specimens have a brown colouration. Little is known about its behaviour. It appears to be inactive and sluggish by day.

Hook-nosed Sea Snake or Beaked sea snake *(Enhydrina schistosa)*

This is a medium-sized snake having a flattened body and tail. Found along the coast and coastal islands. They are seasonally found in the deep sea though they prefer coastal areas. These snakes are generally found in the coast and coastal islands of India. They are amongst the most common of the 20 kinds of sea snakes found in that region. They are active both during the day and at night. They are able to dive up to 100 m and stay underwater for a maximum of five hours before resurfacing. Sea snakes are equipped with glands to eliminate excess salt. They are venomous, but not aggressive and are thus handled by the fishermen without fear, though they are thrown back into the sea upon sight. The venom of this snake is rated four to eight times as toxic as cobra venom. About 1.5 milligrams of its venom is estimated to be lethal. The principal food is fish.

Dwarf Sea snake *(Hydrophis caerulescens)*

Scales on thickest part of the body are quadrangular or hexagonal in shape. There are 14-18 maxillary teeth behind front fangs. Ventrals 253-334 and distinct throughout, though not twice as large as the adjacent body scales. Colour bluish grey above, whitish below, with 40-60 broad bands. This snake ooccurs in Indian Ocean, Coasts of China, South China Sea and Australia.

Class AVES

Class – Aves/birds – 29 orders, 181 families, 8600 species, evolved from reptiles, warm blooded, have feathers, no sweat glands, feet are covered with scales, forelimbs are modified as wings and jaw as beak, no teeth, lay eggs, hallow bones, nest is built to lay eggs, flightless birds – Emu & Ostrich, Birds being feathered bipeds have internal fertilization and lay hard-shelled eggs and are endotherms. Nearly every anatomical feature is related to ability to fly. They are the only animals with feathers that are modified from reptilian scales.

Pelican – Grey Pelican, Dalmatian Pelican and Rosy Pelican –large beak with sack below, large & heavy, eat fish, four toes are webbed, and nest is built on ground, young one fed by regurgitation

Painted Stork –temperate and tropical,large birds, found near lakes & marsh land, prey fish, long neck and legs, toes are webbed, black and white plumage, build nest on trees, diurnal birds

Open Bill Stork – found in wetlands, Gap in mandible, gregarious birds

Ducks – White winged duck & Pink headed duck are the winter migrants from northern hemisphere, Spot billed duck, cotton teal, Comb duck and lesser whistling teal are Indian resident Ducks

Flemingos – America, Africa and Europe, larger Flemingo and lesser Flemingo are found in India (Guj), slender body, large wings, long and thick neck, longer legs, bill is bent in middle, pink plumage, and nest is built by soft mud, large flocks

Vultures – Accipitridae 14 spp & Cathartidae 7 spp, feed on carion, powerful beak for tearing flesh, powerful eyesight and large nest, Bengal vulture, Long billed vulture and Scavenger vulture

Eagles –30 spp, kill its own prey, hooked bills & powerful claws, there is control of rodents & snakes, snatch chicks, Creasted serpent eagle & short toed eagle

Falcons – 60 spp, Caracara- carion eating, True falcon – kills prey in flight & hunt other birds, Shaheen falcon – fastest bird (280 km/hr) & found in hilly terrain, Red headed merleen – hunts flying birds

Jungle Fowl – True pheasents, 4 spp, Ancestral to domestic fowl, male have spurs, fleshy comb, Grey jungle fowl – peninsula, Red jungle fowl – North India

Pea Fowl – have 1.5 mt long feathers with several eyes, flock is of 1 male with 3-6 hens, good eyesight and hearing ability, male have spur on legs and elongated tail feathers, roost on tall trees, come out in late afternoon

Saras Crane – 15 spp, tallest flying birds, found in pairs, close to human habitation

Siberian Crane – USSR, winter migrants of Rajasthan, it is endangered, 5 feet tall, long bills and convoluted wind pipe utter loud call, build nest on ground, monogamous and mate for life

Pigeon – stout body, short leg, small head, short neck, large crop, some are migratory, used to carry messages

Great Indian Bustard–large, long legged, hunted for meat, famous for courting dances, now the bird is protected

Gulls – near water along coastal lines of India, feet are webbed, eat floating garbage, Brown Headed Gull & Black Headed Gull

Terns – Oceanic birds, excellent fliers, Whisked Tern & River Tern are resident, Caspian Tern & Gull Billed Tern are migrant varieties

Parrots – warm climate, mimic human voice, solid green color, large head, short neck and legs, down curved beak, zygodactylis feet, large flock, Rose ringed Parakeet, Large Indian Parakeet

Koel – do not build own nest but lay eggs in others nest, male are black & female are brown, male drives away crows and female lay eggs in crows nest

Owls – predatory birds, nocturnal, forwardly directed eyes gives them stereoscopic vision, acute vision and sharp hearing, barn owls – feed on rats- friends of farmers, Grass owls, jungle owlet, Collared owlet, Great Indian horned owls, Brown Fish Owl

Kingfishers – Common in Tropics, predate fish, White Breasted – give loud ringing call and dive into water, Small Blue Kingfisher – feed on insects

Hornbill – Asia & Africa, Ornamental outgrowth on top plumage, fruit eating, female are kept sealed and fed by male birds, large Indian Pied Hornbill, Malabar Pied Hornbill and Great Indian Hornbill.

Barbets – in group on fruit tree

Wood peckers – feed on wood boring beetles & grubs, has bar tipped tongue, Golden Backed Woodpecker & Mahratta Woodpecker

Drongo – King Crow, Racket Tailed Drongo – forest dweller, Close to lion tailed macaque

Crows – most intelligent birds, Common Crow – close to humans and good scavengers, Jungle Crow – builds nest on trees, feeds on carnivore kills

Mynas – good at imitating human sounds, follow grazing cattle to catch insects, plenty of noise in morning and evening, Hill Myna

Babblers – song birds of tropics, Group of seven so called as seven sisters, continuous chatter hence the name

Weaver bird – male builds the nest by mansoons and female comes decides to mate, female broods and looks after chicks, nest has two entrances one real and other fake to fool snakes

Tailor bird – builds nest by folding the large leaves and sewing the edges together

Exotic Flightless Birds

Ostrich –Struthio camelus, largest bird, 150 kg, Africa, largest egg, legs neck and head are naked, diurnal, live in bands, males make hollow booming call,

Emu – second largest, Australia

Cassowary – New Guinea

Rhea – South America

Kiwi – New Zealand, Nocturnal bird, acute sense of smell

Endemic birds in India are – Himalayan quail, Great Indian Bustard, Forest Owlet, Andaman Serpent Eagle and Narcondom Hornbill

Critically Endangered - Himalayan quail, Great Indian Bustard, Forest Owlet, Siberian Crane, Pink Headed Duck, Red Headed Vulture